SOLAR BATTERY CHARGE CONTROLLER — REV B

Minor Project Report

Submitted by

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B.Tech — PCB Design Project

Software used: KiCad and FreeCAD

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This report describes the circuit and PCB design. The board has not yet been assembled or tested with a real battery.

1. Introduction

Abstract

This project is a solar battery charger designed using KiCad. A solar panel supplies electricity to a PCB, which controls charging of one protected lithium-ion cell. The LT3652 charger converts the higher panel voltage and regulates the charging current and voltage. The final charging-voltage target is 4.2 V and the nominal maximum charging current is 500 mA.

Revision B adds an input fuse, reverse-polarity blocking diode, shutdown jumper and nine test points. We prepared the schematic, PCB layout, 3D view and manufacturing files. Physical charging, LED operation and temperature performance still need hardware testing.

Why this circuit is needed

A nominal 12 V solar panel may operate around 18 V at its maximum-power point. It cannot directly control lithium-ion charging. Our circuit sits between the panel and battery to convert the voltage and control charging. The panel generates electricity, while the battery stores energy.

Objectives

- Design a single-cell solar-charging PCB.
- Set nominal limits of 4.2 V and 500 mA.
- Add input protection and easy measurement access.
- Prepare fabrication files and review the design using CAD checks.

Main specifications

Item	Design target
Battery	One protected 3.7 V nominal lithium-ion cell; 4.2 V charging chemistry
Solar panel	12 V nominal, Vmp about 18 V; cold open-circuit voltage ≤25 V
Charge settings	4.2 V target; 500 mA nominal maximum
Controller	LT3652EMSE
PCB	60 × 55 mm, two copper layers
Temperature sensor	External 10 kΩ NTC, B = 3380 K, attached to the cell

2. Block Diagram and Working

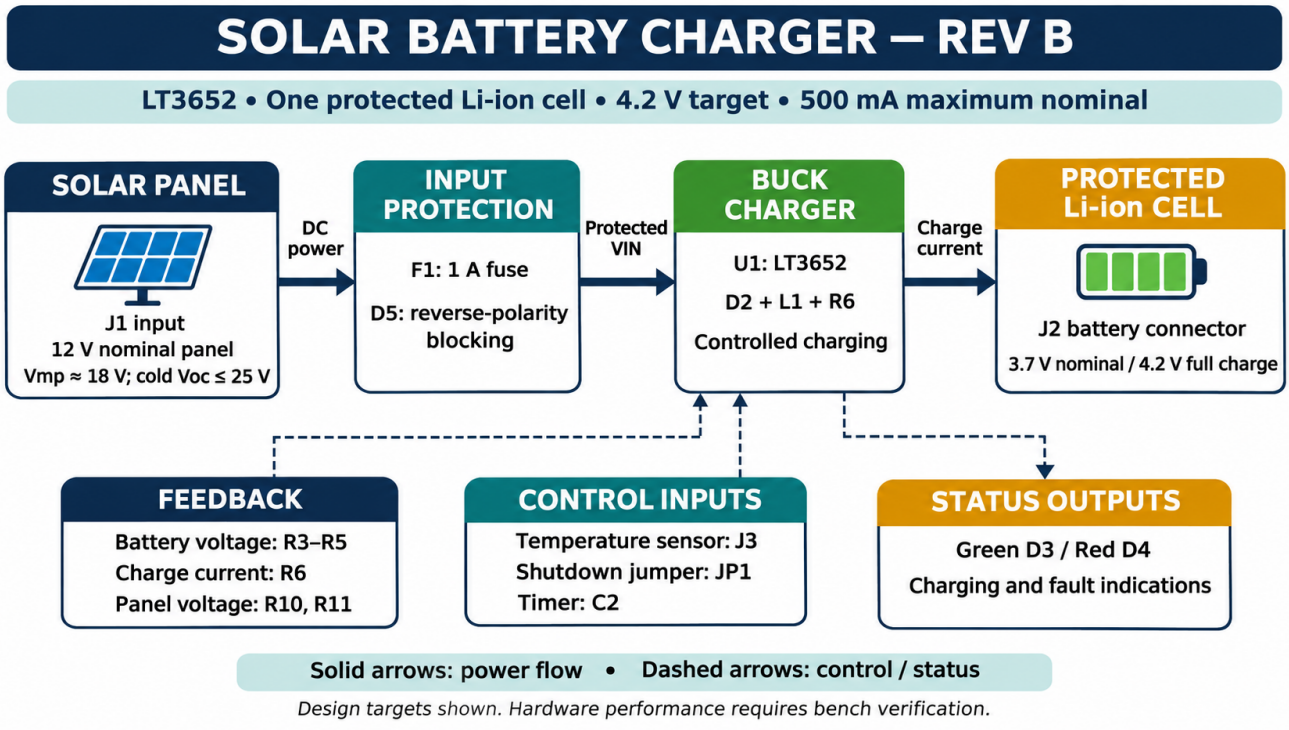


Figure 1: Block diagram of the solar battery charger.

The power flow is: **solar panel → input protection → buck charger → battery.** The feedback circuits monitor voltage and current. The thermistor monitors battery temperature, and the LEDs indicate the charger's status.

Working sequence

1. Sunlight reaches the panel and produces DC electricity.
2. Power enters J1 and passes through the fuse F1 and diode D5.
3. U1 switches power through L1 and R6 towards the battery at J2.
4. R6 helps the controller limit charging current to about 500 mA.
5. Near 4.2 V, the controller holds the voltage near this target while the current decreases.
6. The timer and monitoring circuits control charging and fault responses.

Effect of sunlight

With bright sunlight, more power is available. With weak sunlight, charging slows down or stops. Low sunlight does not automatically drain the battery; stored energy is used when a connected load draws power. This PCB does not include a separate load-management circuit.

3. Schematic Explanation

The schematic has **eight sections**. Matching labels such as VIN, BAT and GND mean those points are connected. The full schematic is attached at the end of this report.

1. Solar input and enable

J1 connects the panel. F1 is an input fuse and D5 blocks reversed panel polarity. C1 and C6 reduce supply disturbances. R7 normally enables charging, while fitting a shunt on JP1 disables it.

2. Charger and power stage

U1 is the LT3652 controller. L1 stores magnetic energy, and D2 provides the freewheel path during switching. R6 senses charging current; D1 and C3 support the switch driver. C4 and C5 filter the output, and J2 connects the battery.

3. Panel-voltage regulation

R10 and R11 monitor the protected input voltage. Near the nominal 17.87 V threshold, U1 reduces charging current to help prevent further panel-voltage collapse. This is fixed input-voltage regulation, not a continuously searching MPPT algorithm.

4. Timer

C2 sets a nominal timer interval of about three hours, with associated preconditioning and fault timing. It does not mean every battery becomes fully charged in three hours.

5. Battery-voltage feedback

R4 and R5 set the final charging-voltage target to 4.2 V. R3 completes the required feedback network. As the battery approaches this voltage, charging current decreases.

6. Temperature sensor

J3 connects an external NTC thermistor attached to the battery. Charging can pause outside the intended temperature range. A disconnected thermistor disables temperature monitoring.

7. Status LEDs

D3 is green and D4 is red. R1 and R2 limit LED current; R8 and R9 pull up the status signals. Green alone indicates charging above about 50 mA; red alone indicates a bad-battery fault; both on indicate a temperature fault. Both off do not prove full charge.

8. Test points

TP1–TP9 provide access to input stages, battery output, current sensing, switching and ground. A 100 mV difference between TP5 and TP6 corresponds to about 500 mA through R6.

4. PCB Design

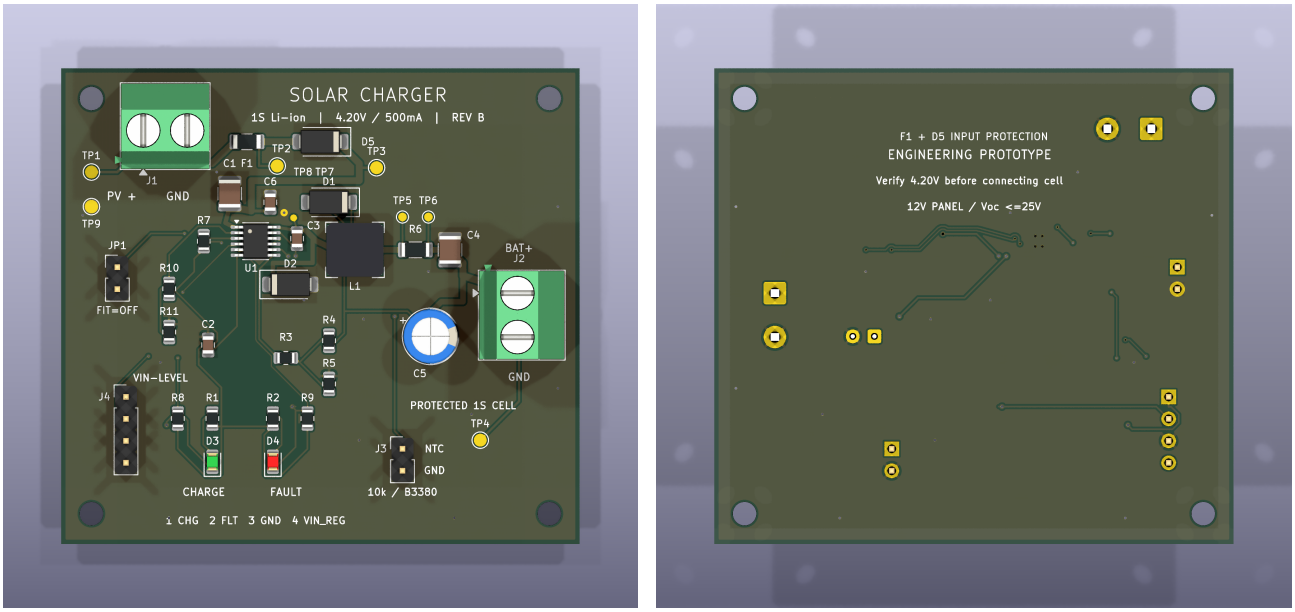


Figure 2: Top and bottom CAD views of Rev B.

Component placement

The panel connector J1 is near the upper-left edge and battery connector J2 is on the right. The protection components are near J1. U1 and the switching components are grouped in the central area. LEDs are near the lower edge, and the jumper, sensor connector and test pads are accessible.

Routing and ground

Copper tracks connect the components according to the schematic. Switching-current paths need compact routing, and current-sense and feedback paths need careful connections to reduce measurement errors. Ground copper and vias provide return paths between the two layers.

Assembly and 3D view

U1 has a grounded exposed pad with four thermal vias to help spread heat into the PCB. Actual temperature must still be measured. Four mounting holes support installation. The 3D view helps inspect component orientation and connector access; some package models are representative.

Keep the local libraries and 3dmodels folder beside the three KiCad files. Open the latest schematic using KiCad 10 or newer. The STEP assembly can also be opened in FreeCAD.

5. Calculations and Components

Main calculations

Charging current: $I = 0.100\ V / 0.20\ \Omega = 0.500\ A$.
Final voltage: $V = 3.3 \times (1 + 60 / 220) = 4.200\ V$.
Input regulation: $V = 2.7 \times (1 + 562 / 100) = 17.874\ V$.
Sense-resistor power: $P = 0.5^2 \times 0.20 = 0.050\ W$.
Timer: $t = 0.68 \times 10^{-6} \times 4.4 \times 10^{6} = 2.992$ hours.

These are nominal calculated values, not measured results. Actual values depend on IC and component tolerances. The input-regulation setting is measured after D5, so the panel-side voltage also includes the diode and wiring drops.

Main component list

Parts	Values / purpose
U1	LT3652EMSE charger controller
J1, J2, J3, J4, JP1	Panel, battery, NTC, status and shutdown connections
F1 / D5	1 A, ≥32 V DC fuse / SS16-class, ≥60 V input diode
L1 / R6	22 µH power inductor / 0.20 Ω, 1%, 0.5 W sense resistor
D1 / D2	SS14-class Schottky diodes for bootstrap and freewheel paths
D3 / D4	Green charging LED / red fault LED
C1 / C6	10 µF / 50 V and 100 nF / 50 V input bypass
C2 / C3	0.68 µF / 10 V timer and 1 µF / 16 V bootstrap
C4 / C5	22 µF / 16 V ceramic and 100 µF / 10 V electrolytic
R1, R2 / R7, R8, R9	10 kΩ LED resistors / 100 kΩ pull-up resistors
R3 / R4 / R5	203 kΩ / 60.0 kΩ / 220 kΩ, all 0.1%, feedback
R10 / R11	562 kΩ / 100 kΩ, input-voltage divider

External items include a protected cell, suitable panel, attached NTC, wiring and appropriate battery-side fusing. Verify actual footprints, ratings and capacitor performance before assembly.

6. Improvements, Checking and Limitations

Difference between Rev A and Rev B

Rev A: Contains the charger, charging/fault LEDs, voltage/current settings and temperature-sensor connection.
Rev B: Retains the 4.2 V / 500 mA targets and adds F1, D5, JP1, nine test points and corrected VIN_REG labeling.

Design checking

The earlier Rev B records show zero ERC and DRC violations, zero unconnected items and zero schematic/PCB mismatches. Its additional scripts recorded 48 circuit checks and 20 layout checks.

For the latest submitted files, a PCB-only DRC found zero violations and zero unconnected pads. All 30 component-model paths and both local library paths resolve. A later layout review passed 19 checks, but its strict “no ignored ERC rules” check did not pass.

The KiCad 10 project has four ERC categories set to ignore: footprint_filter, four_way_junction, simulation_model_issue and single_global_label. These settings need review. Final schematic ERC and schematic/PCB parity must be rerun in KiCad 10; the KiCad 9 environment could not load the newer schematic.

Hardware testing still required

The board has not been physically tested and no electrical simulation is included. First inspect assembly and shorts, then use a current-limited source without a lithium cell. Use a suitable current-sinking battery emulator to check current, voltage, shutdown, temperature faults and charge taper. Test a real protected cell only after the earlier checks pass.

Limitations

- No battery-reversal or input-overvoltage protection.
- A small panel may not provide enough fault current to blow F1.
- Disconnecting the NTC disables temperature monitoring.
- J4 status outputs are VIN-level and not directly MCU-safe.
- No separate load-management stage or dynamic MPPT.
- Efficiency, heating, charging and LED operation are not measured.

Gerber and drill files are included for fabrication review. Supporting folders were reorganized, so some old links and verification scripts need path updates. This does not stop the native KiCad libraries and models from resolving.

7. Individual Contributions and Conclusion

Nishanth R — S20240020316

Schematic and circuit explanation: Study the original circuit and controller functions; explain the eight schematic sections, block diagram, charging flow and setting calculations; review connector functions and Rev A/Rev B differences.

Siddarth Vijayan B — S20240020271

PCB planning and component placement: Review board size, footprints and polarity; explain connector positions, power-stage grouping, capacitor placement, mounting holes and access to LEDs, jumper and test points.

Arvind Kishor R — S20240020270

PCB routing and verification: Explain power/signal routing, ground copper, vias and thermal connections; review design-check results and remaining actions; explain Gerber/drill outputs and the planned testing procedure.

The roles above follow the team’s stated division of work. They do not imply that hardware testing has been completed.

Conclusion

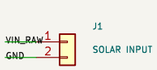
The project provides a PCB design for charging one protected lithium-ion cell from a solar panel. It uses a buck-based charger with nominal limits of 4.2 V and 500 mA. Rev B improves input protection and measurement access. The next steps are final KiCad 10 checks, assembly and controlled bench testing.

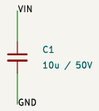
References

1. Team repository: github.com/nishanthr24-proj/PCB-desgning-
2. Original project: github.com/Hirusha785/solar-battery-charge-controller-pcb — README credits Fernando S.M.H.G.
3. Analog Devices, LT3652 datasheet, document 3652fe.
4. KiCad documentation and libraries: docs.kicad.org and KiCad packages3D.
5. FreeCAD documentation: freecad.org.

Original-source and library credits are retained in ATTRIBUTION.md and licenses. AI tools assisted CAD preparation and report drafting. Full schematic reference drawing follows on the last page.

1. SOLAR INPUT / ENABLE

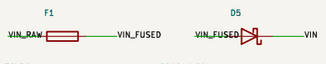


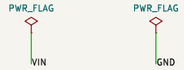




Panel: 12 V nominal, Vmp ~18 V
Cold Voc <=25 V. F1/D5 protect solar input.
Battery reversal and overvoltage are NOT protected.

FUSED REVERSE-POLARITY SOLAR INPUT







2. LT3652 CHARGER / POWER STAGE

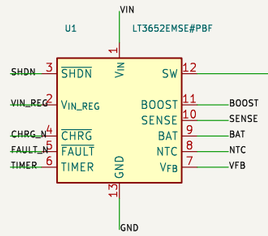


JP1 OPEN = enabled; fit shunt = shutdown.
R7 limits pull-up current; do not short VIN to GND.

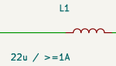


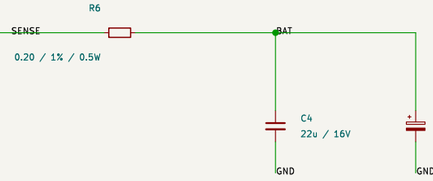












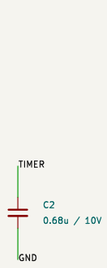
R6 sets 500 mA maximum charge current.
1S protected Li-ion cell only: 4.20 V full charge.
Battery connector pin 1 = BAT+; pin 2 = GND.

3. PANEL VOLTAGE REGULATION



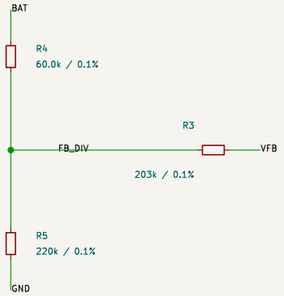
Protected VIN regulation target = 17.874 V.
Panel target = 17.874 V + D5 drop.
Fixed voltage regulation; not dynamic MPPT.

4. TIMER



Nominal 3 h; bad-battery
preconditioning timeout ~22.5 min.
Timer retries if float not reached.

5. PRECISION BATTERY FEEDBACK



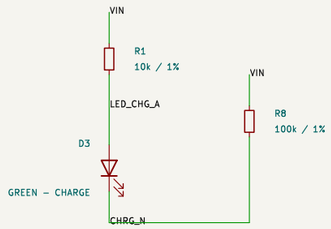
Vfloat = 3.3*(1+60.0k/220k) = 4.200 V
R3 + (R4 || R5) = 250.14 kohm
Use specified 0.1% feedback resistors.

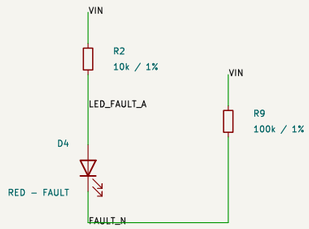
6. TEMPERATURE SENSOR



Attach 10k NTC, B=3380 K, to cell.
Nominal charge window ~0 to 40 C.
Open sensor DISABLES temperature monitoring.

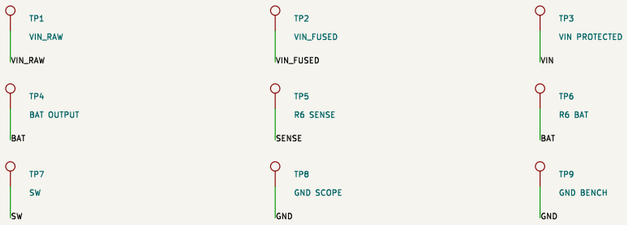
7. ADDED STATUS LEDs





Green only: charging >=C/10 (50 mA nominal).
Red only: bad-battery timeout. Both: NTC fault.
Both off: standby, no input, or low charging current.
LEDs are powered from solar VIN; off does not prove full charge.

8. REV B PROBE POINTS (see test plan)





VIN-LEVEL STATUS
NOT MCU SAFE
VIN_REG: analog sense only

ENGINEERING PROTOTYPE - ERC/DRC do not replace bench verification. No lithium cell until dummy-load checks pass.

Based on the uploaded Fernando S.M.H.G. project; corrected circuit and layout
1S Li-ion / 4.20 V / 500 mA - engineering prototype

Sheet: /
File: Solar_Charger_LED_RevB.kicad_sch

Title: Solar Charger Rev B - Fused Input, Shutdown and Test Points

Size: A3	Date: 2026-11-06	**Rev: B**
KiCad E.D.A. 9.0.2		Id: 1/1